

MINI PROJECT COMPLETION REPORT

Submitted By

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INDUSTRIAL MACHINE HEALTH MONITORING USING MQTT

Mentored by

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INTRODUCTION

Industrial machines are widely used in manufacturing and production industries. The continuous operation of these machines is essential for maintaining productivity and reducing losses. In this project, IoT technology and the MQTT protocol are used to monitor machine parameters such as voltage, current, and vibration in real time. This helps in identifying faults at an early stage and improves machine reliability.

PROBLEM STATEMENT

Industrial machines may develop faults unexpectedly, leading to production downtime and maintenance expenses. Traditional monitoring methods do not provide continuous real-time machine status. Hence, a reliable monitoring system is needed to detect abnormal conditions at an early stage.

NEED FOR THE PROJECT

Continuous monitoring of machine health helps prevent sudden breakdowns and improves operational efficiency. Real-time data collection enables timely maintenance and fault detection. This project reduces downtime, maintenance costs, and production losses.

OBJECTIVES

1. To monitor machine parameters such as voltage, current, and vibration continuously.
2. To transmit machine data using the MQTT communication protocol.
3. To identify abnormal machine conditions at an early stage.
4. To provide fault indication and machine protection during abnormal conditions.
5. To improve machine reliability and reduce maintenance costs.

EXISTING SYSTEM

Machine health monitoring plays an important role in industries for preventing equipment failures and improving operational efficiency. Various conventional monitoring methods such as manual inspection, meter-based monitoring, and periodic maintenance are commonly used. These methods help in assessing machine conditions but often require human involvement.

PROPOSED METHODOLOGY

The proposed Industrial Machine Health Monitoring System uses IoT technology and the MQTT communication protocol to monitor the condition of industrial machines in real time.

The system continuously measures important machine parameters such as voltage, current, and vibration using appropriate sensors. The collected sensor data is sent to the ESP32 microcontroller, which acts as the central processing unit of the system.

The ESP32 reads and processes the sensor values and compares them with predefined threshold limits. The processed information is then transmitted to an MQTT broker through a Wi-Fi connection. Users can monitor the machine status remotely using an MQTT dashboard or mobile application.

When abnormal conditions such as the over current is detected, the system immediately generates an alert. The RGB LED changes its status indication, the buzzer provides an audible warning, and the relay disconnects the machine supply to prevent damage. This enables early fault detection and improves machine safety.

Hardware Used

- ESP32 Microcontroller
- Voltage Sensor
- Current Sensor
- Vibration Sensor
- 2-Channel Relay Module
- RGB LED
- Buzzer
- 12V DC Motor
- 12V DC Power Supply

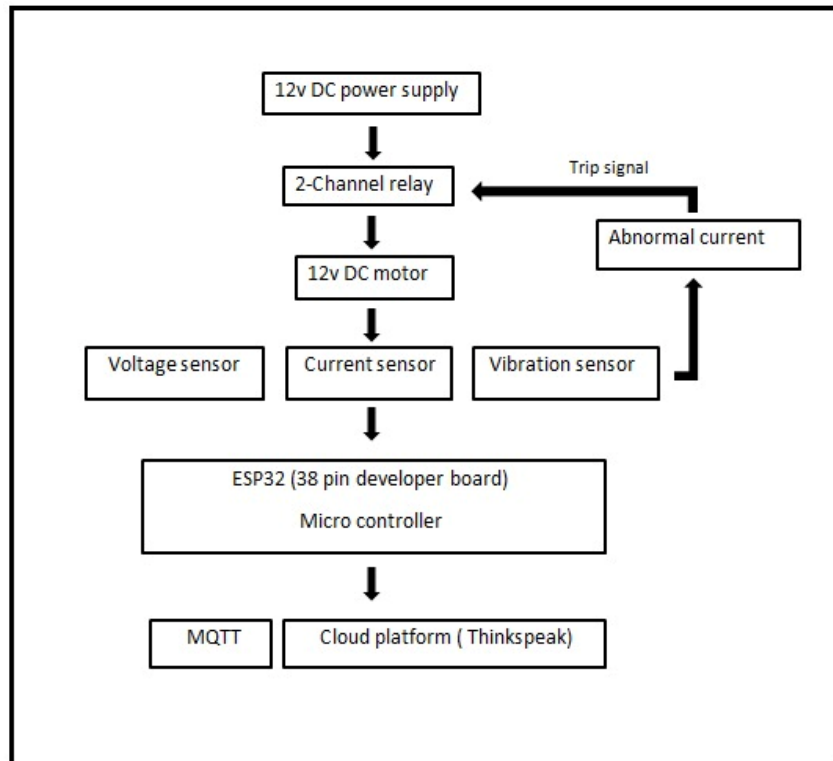
Software Used

- Arduino IDE
- MQTT Protocol
- Thingspeak

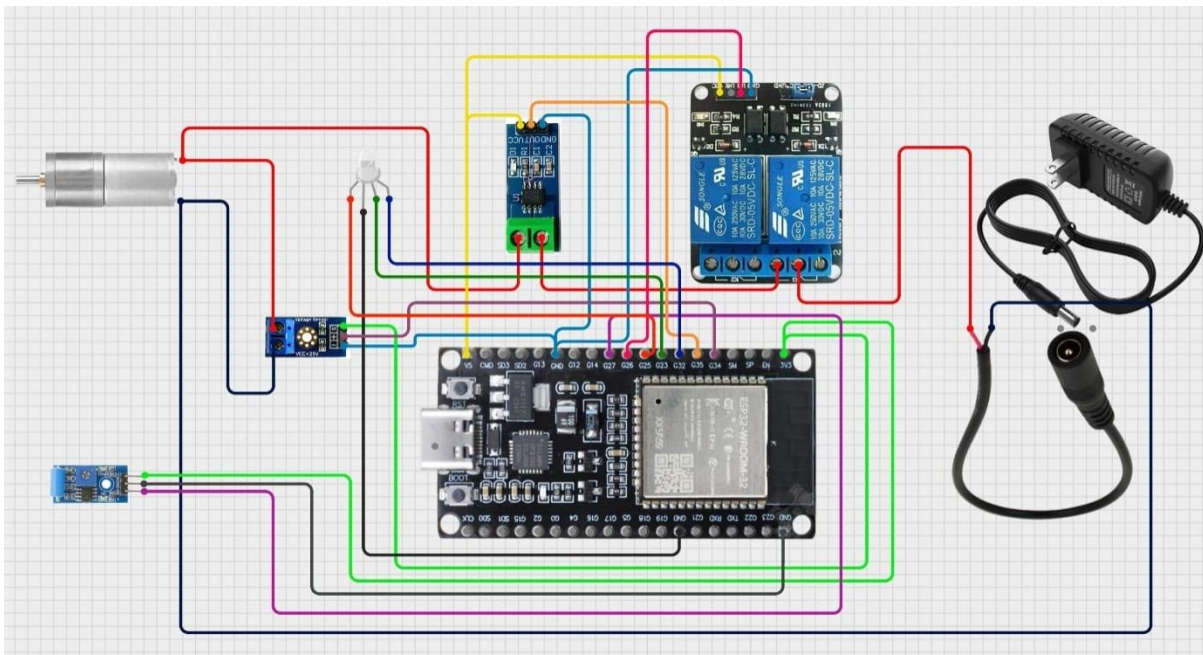
Input : The input to the system is obtained from voltage, current, and vibration sensors installed on the machine.

Processing : The ESP32 receives sensor data, processes the values, checks for abnormal conditions, and publishes the information through the MQTT protocol and Thingspeak cloud platform.

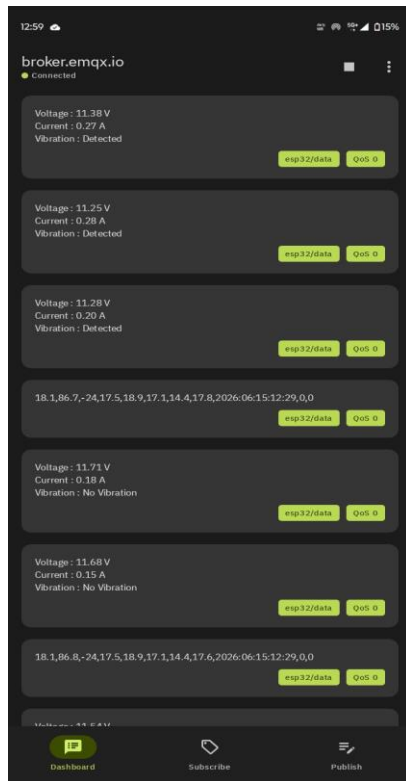
Block Diagram



Software Design (Circuit diagram)



Output : The machine status is displayed on the dashboard (MQTT and Thingspeak). During fault conditions, warning indications are provided through LEDs and a buzzer, and the relay trips the machine for protection.



MQTT Dashboard



Thingspeak

Applications:

1. Industrial machine condition monitoring.
2. Factory equipment health monitoring.
3. Predictive maintenance systems.
4. Remote monitoring of motors and machines.
5. Small and medium-scale industries for fault detection.

Future Scope:

1. Real time implementation in AC Machines
2. Cloud-based data storage and analysis
3. Mobile application for real-time monitoring.
4. Monitoring multiple machines using a single MQTT server.
5. Automatic maintenance alerts and report generation.

